Detecting Heart Disease: A Comparative Statistical Analysis of Clinical Markers

Course
Data 602 Statistical Data Analysis

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1. Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) is a leading cause of death worldwide (WHO, 2021). Early detection using accessible clinical markers, such as blood panels and exercise stress tests, can help prioritize patients for invasive diagnostics and inform risk-stratification tools. This project aims to:

- Identify which simple clinical measures (e.g., cholesterol, exercise test results) best distinguish p atients who should be prioritized for further testing.
- Build and evaluate interpretable predictive models that could support risk stratification and diagnostic decision-making.
- Translate the findings into clear, evidence-based information for patients about which health markers matter most.

1.2 Research Objectives and Questions

Our primary goal is to determine which clinical measurements are significantly associated with heart disease. We evaluated three specific research questions:

- **Difference of means (thalch):** Is there a significant difference in mean maximum heart rate between patients with and without heart disease?
- **Difference of means (chol):** Is there a significant difference in mean serum cholesterol between the two groups?
- **Difference of proportions (exang):** Is there a significant difference in the proportion of patients experiencing exercise-induced angina?

We address these using conventional inference methods (two-sample t-tests and two-proportion tests) p aired with 5,000-resample bootstrap confidence intervals.

2. Data and Variables

2.1 Data Source and Study Population

This retrospective, observational study uses a 16-attribute subset of the 1988 UCI Heart Disease dataset sourced via Kaggle (Janosi et al., 1989; Karim, n.d.). The data consists of existing medical records collected between 1981 and 1984 from adult patients evaluated for suspected coronary artery disease (Detrano et al., 1989). Treated at the Cleveland Clinic, the Hungarian Institute of Cardiology, University Hospitals in Switzerland, and the VA Medical Center in Long Beach, these patients represent a convenience sample (Stallard, 2026). To protect privacy, the original researchers replaced all identifying information with dummy values before compiling it. (Janosi et al., 1989).

2.2 Variables of Interest

We are analyzing three clinical predictor variables and one target diagnosis variable:

- thalch: The maximum heart rate achieved by the patient during exercise testing (numeric).
- chol: The patient's serum cholesterol level measured in mg/dl (numeric).
- exang: Whether the patient experienced exercise-induced angina, recorded as True or False (categorical).
- num: The baseline heart disease diagnosis. This was originally recorded as 0 for no disease and 1 to 4 for increasing severity (Janosi et al., 1989). We converted this into a binary target variable named “diseased”, where 0 indicates no disease, and 1 indicates the presence of heart disease.

2.3 Data Cleaning and Preprocessing

We loaded the data into R directly from Kaggle using the RKaggle package and made the following changes before analysing it.

Fixing impossible values
Some patients had a cholesterol (chol) or resting blood pressure (trestbps) value of 0. These values are not possible in a living patient, so they are placeholders for missing data. We changed them to NA so they would not distort our results.
Creating a disease indicator
The original diagnosis variable, num, is 0 for no disease and 1 to 4 for different levels of disease. Our questions only compare patients with and without heart disease, so we created a new variable, “diseased”. It equals 1 if num is greater than 0 and 0 otherwise.

Renaming a variable
We renamed the dataset to origin, since it shows which of the four hospitals each patient came from.

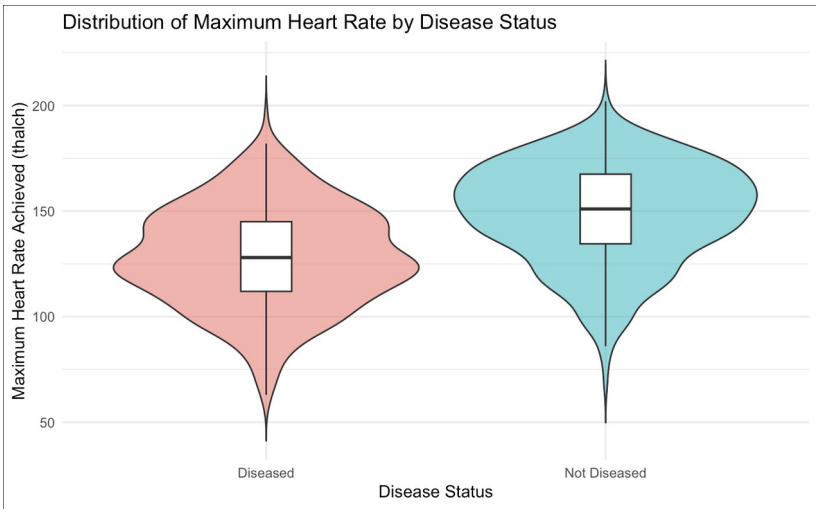
3. Difference in Mean Maximum Heart Rate (thalch)

3.1 Methods
H_0: μ_Diseased = μ_Not Diseased (the mean maximum heart rate is the same in both groups)
H_a: μ_Diseased ≠ μ_Not Diseased (the mean maximum heart rate differs between the groups)

We compared the mean maximum heart rate between patients with and without heart disease u sing a pooled two-sample t-test, which gives a 95% confidence interval for the difference in means (Diseased − Not Diseased). We also built a bootstrap confidence interval for the same difference by resampling each group with replacement 5,000 times and taking the 2.5th and 97.5th percentiles of the resampled mean differences, then compared it with the t-based interval.

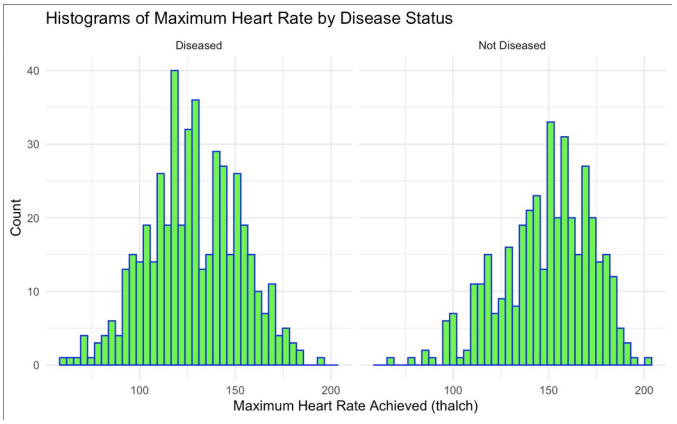
Variance Check

We chose the pooled t-test because the violin and boxplots show nearly identical spread in both groups (IQRs of about 33 bpm each, with similar whisker lengths and violin shapes), and the sample standard deviations were nearly equal (24.02 bpm for Diseased and 23.61 bpm for Not Diseased, ratio = 1.02). The equal-variance assumption is therefore r easonable.



Normality Check





Normal QQ plots and histograms of maximum heart rate were examined for each group. In the QQ plots, the points follow the reference line closely in both groups, with only minor departures in the tails. The Diseased group looks approximately normal, and the Not Diseased group shows mild left skew, which the histograms also show as a longer lower tail. Small bumps in the histograms are likely due to bin width rather than real subgroups. Because both groups are large (474 diseased and 391 not diseased patients), the sampling distribution of the mean difference should be approximately normal, so the t-test assumptions are reasonably satisfied. The bootstrap interval in the later section provides a further check on this.

3.2 Results

Two-Sample t-test

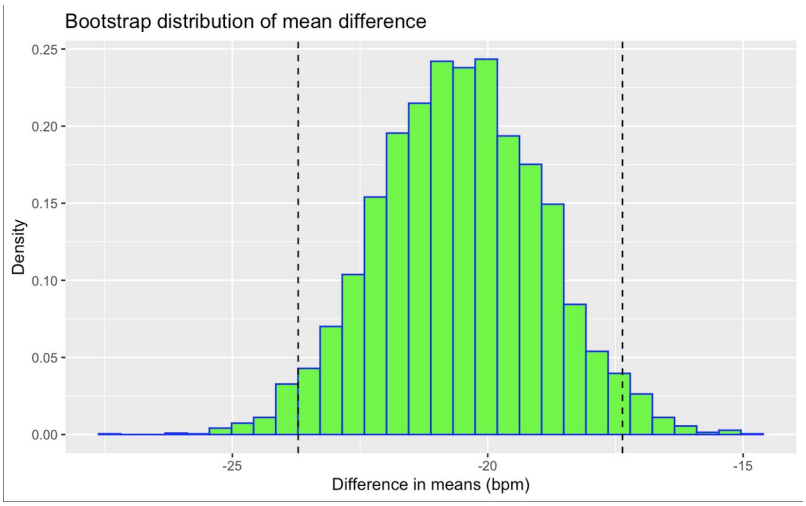
People with heart disease reached a mean maximum heart rate of about 128.3 bpm, compared with about 148.8 bpm for those without. The estimated difference (Diseased − Not Diseased) is −20.5 bpm.

```
        Two Sample t-test

data:  thalch by disease_status
t = -12.612, df = 863, p-value < 0.00000000000000022
alternative hypothesis: true difference in means between group Diseased and group Not Diseased is not equal to 0
95 percent confidence interval:
 -23.73519 -17.34262
sample estimates:
    mean in group Diseased mean in group Not Diseased
                  128.2616                  148.8005
```

Confidence interval: We are 95% confident that the true mean difference lies between −23.7 and −17.3 bpm. Since the interval is entirely below zero, the mean maximum heart rate is plausibly 1 7 to 24 bpm lower in the diseased group.

Test statistic and p-value: t = −12.61 on 863 degrees of freedom, with p < 2.2e-16. If there were truly no difference in means, a gap this large or larger would be extremely unlikely if there were no true difference, so we reject the null h ypothesis of equal means.



Bootstrap Confidence Interval

The bootstrap distribution of the mean difference is approximately bell-shaped and centered near −20.5 b pm. The dashed vertical lines mark

the 2.5th and 97.5th percentiles, which form the 95% bootstrap percentile interval of −23.71 to −17.36 bpm.

method <chr>	lower <dbl>	upper <dbl>
Pooled t	−23.73519	−17.34262
Bootstrap	−23.71141	−17.36233

Table 1. 95% confidence intervals for the difference in mean maximum heart rate (Diseased − Not Diseased), in bpm

We compared the mean maximum heart rate (thalch) of patients with and without heart disease. The pooled two-sample t-test gave a 95% confidence interval of −23.74 to −17.34 bpm for the difference in means (Diseased − Not Diseased). The bootstrap percentile interval, based on 5,000 r esamples, was −23.71 to −17.36 bpm, nearly identical to the conventional interval.

Since the bootstrap does not rely on normality, this agreement supports the view that the t-test assumptions were reasonably met, despite mild left skew in the Not Diseased group. Both intervals exclude 0, so we conclude that patients with heart disease reach a lower maximum heart rate, by about 20.5 bpm on average (plausibly 17 to 24 bpm). This fits the idea that a diseased heart has a reduced ability to speed up under exertion (chronotropic incompetence), and it supports thalch as a useful indicator when screening for heart disease.

4. Difference in Mean Cholesterol (chol)

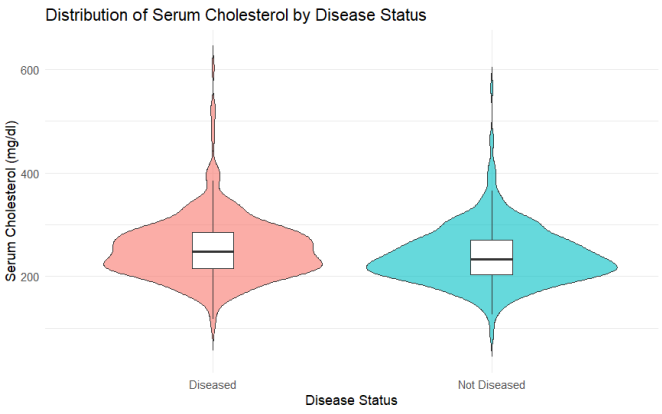
4.1 Methods

Let μ_{Diseased} be the true mean serum cholesterol (mg/dl) of patients with heart disease, and $\mu_{\text{Not Diseased}}$ the true mean serum cholesterol (mg/dl) of patients without heart disease, in the population of patients evaluated for suspected coronary artery disease at these hospitals.

H_0: μ_Diseased = μ_Not Diseased (the mean serum cholesterol is the same in both groups)

H_a: μ_Diseased ≠ μ_Not Diseased (the mean serum cholesterol differs between the groups)

As in Section 3, we compared the two groups using a pooled two-sample t-test, which gives a 95% confidence interval for the difference in means (Diseased − Not Diseased), and a 95% bootstrap p ercentile interval based on 5,000 resamples of each group. Of the 920 patients, 202 (22%) had no usable cholesterol value (a recorded value of 0 or a missing entry), so the analysis uses 346 diseased and 372 not diseased patients. All 123 patients from Switzerland are missing cholesterol, so that hospital is not represented in this section.

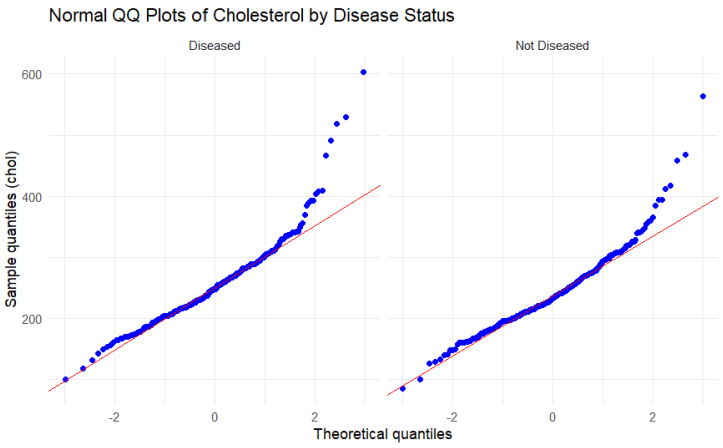


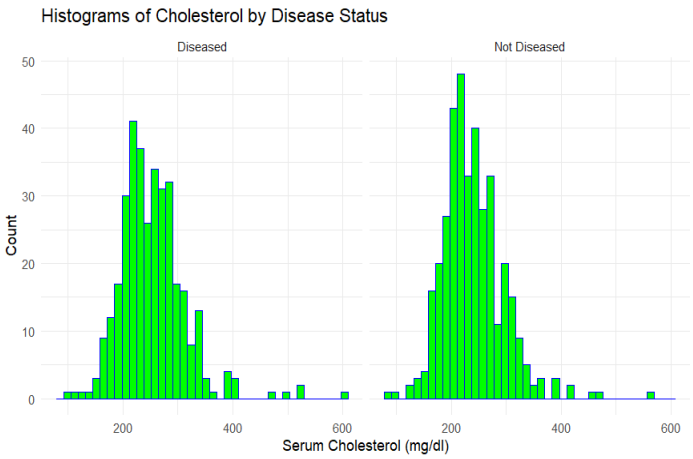
Variance Check

The violin and boxplots show similar spread in both groups. The sample standard deviations were 60.6 mg/dl for Diseased and 55.8 mg/dl for

Not Diseased (ratio = 1.09), so the equal-variance assumption is reasonable and we used the pooled test.

Normality Check





Both groups are right-skewed: the upper tails of the QQ plots curve above the reference line, and the histograms have a longer upper tail with a few very high readings (the largest is 603 mg/dl). Because both groups are large (346 and 372 patients), the sampling distribution of the mean difference should still be approximately normal. The bootstrap interval, which does not rely on normality, provides a further check. If the two intervals disagreed, we would rely on the bootstrap interval.

Independence cannot be confirmed because the patients are a convenience sample from three hospitals rather than a random sample, so the results describe these patients and should not be generalized beyond them.

4.2 Results

Two-Sample t-test

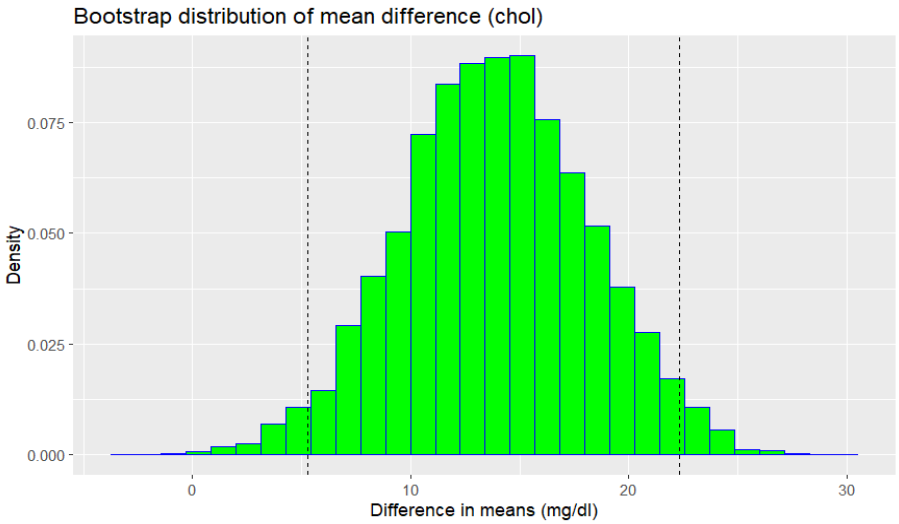
```
        Two Sample t-test

data:  chol by disease_status
t = 3.1886, df = 716, p-value = 0.001492
alternative hypothesis: true difference in means between group Diseased and group Not Diseased is not equal to 0
95 percent confidence interval:
  5.322337 22.377800
sample estimates:
    mean in group Diseased mean in group Not Diseased
                  254.0087                  240.1586
```

Patients with heart disease had a mean cholesterol of 254.0 mg/dl, compared with 240.2 mg/dl for those without. The estimated difference (Diseased − Not Diseased) is +13.8 mg/dl. The test statistic is t = 3.19 on 716 degrees of freedom, with p = 0.0015. If there were truly no difference in means, a gap this large would be unlikely to occur by chance, so we reject the null hypothesis of equal means. We are 95% confident that the true mean difference lies between 5.3 and 22.4 mg/dl.

Bootstrap Confidence Interval

The bootstrap distribution of the mean difference is approximately bell-shaped and c entered near +13.8 mg/dl. The 9 5% bootstrap percentile interval i s 5.2695 to 22.3391 mg/dl, nearly identical to the t-based interval (Table 2).



Description: df [2 × 3]

	method <chr>	lower <dbl>	upper <dbl>
	Pooled t	5.322337	22.37780
2.5%	Bootstrap	5.269471	22.33907

2 rows

Table 2. 95% confidence intervals for the difference in mean cholesterol (Diseased − Not Diseased), in mg/dl

Because the two intervals agree, the right skew does not appear to have affected the t-test, and both intervals exclude 0. We conclude that patients with heart disease in this sample have higher serum cholesterol on average, by plausibly 5 to 22 mg/dl. The difference had the same direction at each of the t hree hospitals with data (Cleveland +9.5, Hungary +28.6, VA Long Beach +12.5 mg/dl), so the pooled result is not driven by a single site.

The difference is small relative to the spread within each group (standard deviations of about 56 to 61 m g/dl), so the two groups overlap heavily. Cholesterol differs between the groups on average, but on its o wn it is a weak indicator for screening.

Missing Data by Hospital

A tibble: 4 × 4

origin <chr>	n <int>	n_missing_chol <int>	pct_diseased <dbl>
Cleveland	304	0	45.7
Hungary	293	23	36.2
Switzerland	123	123	93.5
VA Long Beach	200	56	74.5

Missing cholesterol is concentrated at one hospital. All 123 Switzerland patients are missing, and Switzerland has a very high disease rate (93%, compared with 36–46% at Cleveland and Hungary). Removing these patients therefore changes the mix of the sample, and our results apply only to the other hospitals.

5. Difference in Proportions of Exercise-Induced Angina (exang)

5.1 Methods

Let p_diseased be the true proportion of patients with heart disease who experience exercise-induced angina, and p_not diseased the true proportion of patients without heart disease who experience it, in the population of patients evaluated for suspected coronary artery disease at these hospitals.

H0: p_diseased = p_not diseased (the proportion with exercise-induced angina is the same in both groups)
Ha: p_diseased ≠ p_not diseased (the proportion with exercise-induced angina differs between the groups)

Exercise-induced angina is a yes/no variable, so instead of comparing means we compare the proportion of patients with angina in each group. Since the two groups are independent and both are large, we used a two-proportion test with prop.test() in R, which gives the p-value and a 95% confidence interval for the difference in proportions (Diseased − Not Diseased). prop.test() reports a chi-square statistic, which for a comparison of two proportions equals z^2, so z is its square root.

We also built a bootstrap percentile interval by resampling each group with replacement 5,000 times and taking the 2.5th and 97.5th percentiles of the resampled differences, then compared it with the prop.test() interval.

Of the 920 patients, 55 had no recorded exang value, so the analysis uses 474 diseased and 391 not diseased patients. Most of the missing values come from VA Long Beach (53 of its 200 patients). Hungary and Switzerland are each missing one, and Cleveland is missing none.

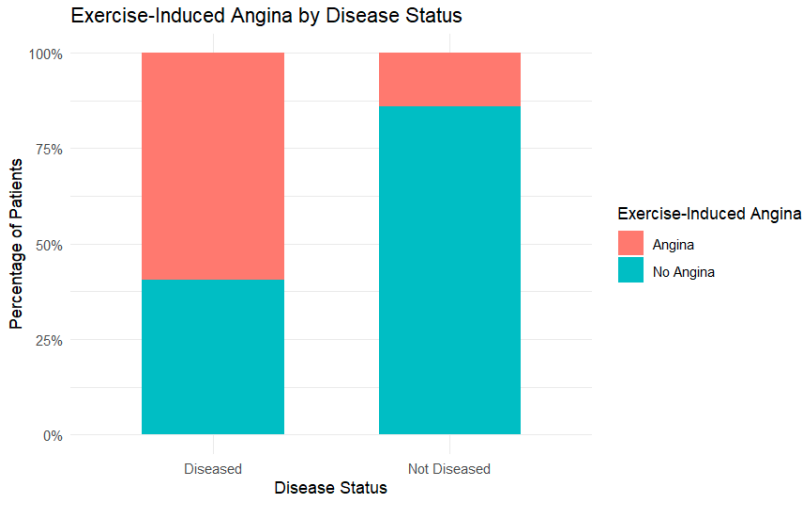
origin <chr>	n <int>	n_missing_exang <int>	pct_diseased <dbl>
Cleveland	304	0	45.7
Hungary	293	1	36.2
Switzerland	123	1	93.5
VA Long Beach	200	53	74.5

4 rows

	Angina	No Angina
Diseased	282	192
Not Diseased	55	336

The bar chart shows the same counts as percentages. Angina makes up most of the diseased bar but only a small part of the not diseased bar.

Independence cannot be confirmed because the patients are a convenience sample rather than a random sample, so the results describe these patients and s hould not be generalized beyond them.



5.2 Results

Two-Proportion Test

About 59.5% of patients with heart disease had exercise-induced angina, compared with 14.1% of those without, a difference of 45.4 percentage points. The test gave X-squared = 185.93 on 1 degree of freedom (z = 13.64), with p approx 0. If there were truly no difference in proportions, a gap this large would be essentially impossible by chance, so we reject the null hypothesis. We are 95% confident that t he true percentage of patients with angina is between 39.8 and 51.0 percentage points higher in the diseased group than in the not diseased group.

Bootstrap Confidence Interval

The bootstrap distribution of the difference in proportions is approximately bell-shaped and centred near 45.4 percentage points. The dashed vertical lines mark the 2.5th and 97.5th percentiles, which form the 95% bootstrap percentile interval of 39.7 to 5 1.1 percentage points.

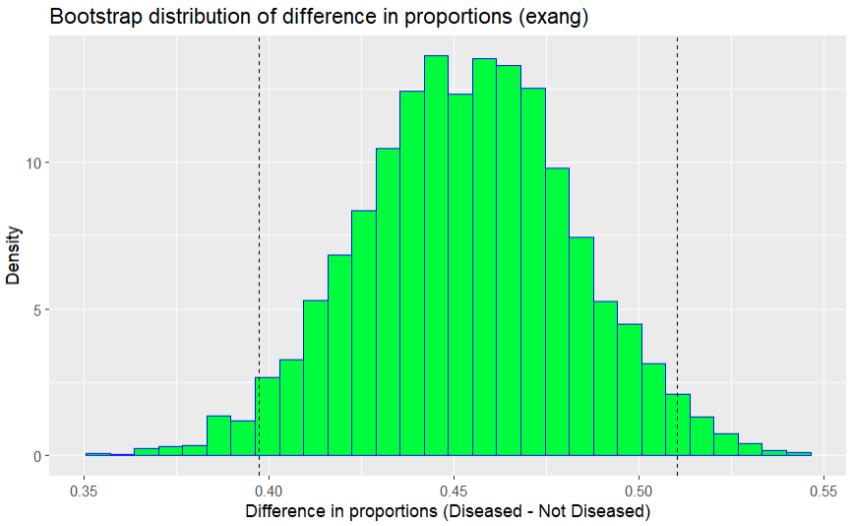
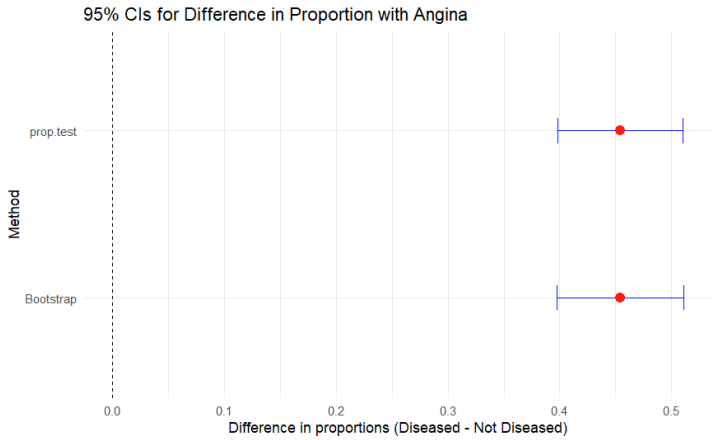


Table 3. 95% confidence intervals f or the difference in proportion with exercise-induced angina (Diseased − Not Diseased), in percentage points:

Method	Estimate	Lower	Upper
prop.test	45.4	39.8	51.0
Bootstrap	45.4	39.7	51.1



The dot-and-whisker plot shows the two intervals side by side, and both sit well to the right of the dashed line at zero. Because the bootstrap does not rely on the normal approximation, this agreement supports the prop.test() result. Both intervals exclude 0, so we conclude that exercise-induced angina is more common among patients with heart disease in this sample, by plausibly 40 to 51 percentage points. In relative terms, it occurred about four times as often in the diseased group ($0.595 / 0.141 \approx 4.2$).

6. Limitations

This analysis is observational, so our results show associations, not causation. Factors we did not adjust for, such as age, sex, fasting blood sugar level, and number of major vessels colored by fluoroscopy, may partly explain the differences between the groups. The data were collected at four hospitals in the early 1980s, so the results may not fully apply to patients today or to other populations. We also combined all disease severity levels (num 1 to 4) into a single "diseased" group, which hides possible differences between mild and severe cases. Finally, some values were missing or recorded as impossible zeros, and we excluded these from the relevant analyses, which could introduce bias if the values were not missing at random.

7. Conclusion

This comparative analysis evaluated the diagnostic utility of thalch, chol, and exang across 920 cardiac patients. Our dual-methodology approach supports that conventional parametric methods remained highly robust, as t-test and proportion intervals were virtually identical to 5,000-iteration bootstrap intervals across all markers.

While all three markers showed statistically significant differences, dynamic responses to exercise stress tests (thalch and exang) provided substantially greater diagnostic separation than static blood markers. Diseased patients averaged a 20.5 bpm deficit in peak heart rate and experienced exercise-induced angina over four times as often. Conversely, heavy group overlap limits cholesterol's utility as an isolated screening tool. In clinical practice, prioritizing exercise stress tests provides immediate, high-contrast indicators for triaging cardiovascular care.

Appendix

Appendix A - References

Detrano, R., Jánosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J., Sandhu, S., Guppy, K., Lee, S., & Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *American Journal of Cardiology*, 64(5), 304-310.

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). *Heart Disease* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52P4X

Karim, R. (n.d.). *Heart Disease Data* [Dataset]. Kaggle. Retrieved from https://www.kaggle.com/datasets/redwankarimsony/heart-disease-data/data

Stallard, J. (2026). *DATA 602 Statistical Data Analysis: Course Notes*. University of Calgary.

World Health Organization. (2021). *Cardiovascular diseases (CVDs)*. Retrieved from https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

Appendix B - Code

1. Data Cleaning and Preparation Code

```{r}
install.packages("RKaggle", Ncpus=4, quiet=TRUE)
#please run this to install RKaggle Library so the dataset can be imported  at your end
```

```{r}
# Load packages
library(RKaggle)
library(readr)
library(dplyr)
library(ggplot2)
library(mosaic)
```

```{r}
health_data_df = get_dataset("redwankarimsony/heart-disease-data")
```

```{r}
# Data cleaning

health_data_df = health_data_df %>% # sets impossible values to NA
  mutate(
    chol = ifelse(chol == 0, NA, chol),
    trestbps = ifelse(trestbps == 0, NA, trestbps),
    diseased = ifelse(num != 0, 1, 0) # creates a binary disease indicator
  ) %>%
  rename(origin = dataset) # renames "dataset"

head(health_data_df, 4)
```

2. Statistical Analysis

Maximum Heart Rate (Thalch):

```{r}
thalch_df = health_data_df %>%
  filter(!is.na(thalch)) %>%
  mutate(disease_status = ifelse(diseased == 1, "Diseased", "Not Diseased"))
```

```{r}
ggplot(thalch_df, aes(x = disease_status, y = thalch, fill = disease_status)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.15, fill = "white", outlier.shape = NA) +
  labs(x = "Disease Status",
       y = "Maximum Heart Rate Achieved (thalch)",
       title = "Distribution of Maximum Heart Rate by Disease Status") +
  theme_minimal() +
  theme(legend.position = "none")
```

```{r}
# BootStrap
N_bootstraps = 5000
boot_diff <- numeric(N_bootstraps)

x = thalch_df$talch[talch_df$disease_status == "Diseased"]
y = thalch_df$talch[talch_df$disease_status == "Not Diseased"]

set.seed(42)
for (i in 1:N_bootstraps) {
  x_sample = sample(x, length(x), replace = TRUE)
  y_sample = sample(y, length(y), replace = TRUE)
  boot_diff[i] <- mean(x_sample) - mean(y_sample)
}

boot_diff_df = data.frame(boot_diff)
```

```{r}
sd_diseased = sd(x)
sd_not_diseased = sd(y)
sd_ratio <- max(sd_diseased, sd_not_diseased) / min(sd_diseased, sd_not_diseased)

round(c(sd_diseased, sd_not_diseased, sd_ratio), 2)
```

```{r}
# QQ Plot
ggplot(thalch_df, aes(sample = thalch)) +
  stat_qq(col="blue") +
  stat_qq_line(col="red") +
  facet_wrap(~disease_status) +
  theme_minimal() +
  labs(title = "Normal QQ Plots of Maximum Heart Rate by Disease Status",
       x = "Theoretical quantiles",
```

```
      y = "Sample quantiles (thalch)")
```
```{r}
ggplot(thalch_df, aes(x = thalch)) +
  geom_histogram(bins = 40, fill = "green", colour = "blue") +
  facet_wrap(~disease_status) +
  theme_minimal() +
  labs(title = "Histograms of Maximum Heart Rate by Disease Status",
       x = "Maximum Heart Rate Achieved (thalch)",
       y = "Count")
```

```{r}
# Two-sample t-test
t.test(thalch ~ disease_status, data = thalch_df, var.equal = TRUE)
```
```{r}
# 95% percentile bootstrap CI (Diseased - Not Diseased)
boot_ci <- qdata(~boot_diff, c(0.025, 0.975), data=boot_diff_df)
boot_ci
```
```{r}
# bootstrap distribution
ggplot(boot_diff_df, aes(x = boot_diff, y = after_stat(density))) +
  geom_histogram(fill = "green", col = "blue") +
  geom_vline(xintercept = boot_ci, linetype = "dashed") +
  labs(title = "Bootstrap distribution of mean difference",
       x = "Difference in means (bpm)", y = "Density")
```
```{r}
# pooled t-test and Bootstrap interval
t_ci <- t.test(x, y, var.equal = TRUE)$conf.int
data.frame(method = c("Pooled t", "Bootstrap"),
           lower = c(t_ci[1], boot_ci[1]),
           upper = c(t_ci[2], boot_ci[2]))
```

Cholesterol (Chol):

```{r}
chol_df = health_data_df %>%
  filter(!is.na(chol)) %>%
  mutate(disease_status = ifelse(diseased == 1, "Diseased", "Not Diseased"))

# Missing cholesterol by hospital (for Section 2.4 / Limitations)
chol_missing = health_data_df %>%
  group_by(origin) %>%
  summarise(n = n(),
            n_missing_chol = sum(is.na(chol)),
            pct_diseased = round(100 * mean(diseased), 1))
chol_missing
```

```

```{r}
# Descriptive statistics by group
chol_df %>%
  group_by(disease_status) %>%
  summarise(n = n(), mean = mean(chol), sd = sd(chol),
            median = median(chol), max = max(chol))
```
```{r}
# Mean cholesterol by hospital and group (same direction at each hospital?)
chol_df %>%
  group_by(origin, disease_status) %>%
  summarise(n = n(), mean = round(mean(chol), 1), .groups = "drop")
```
```{r}
# Variance check
ggplot(chol_df, aes(x = disease_status, y = chol, fill = disease_status)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.15, fill = "white", outlier.shape = NA) +
  labs(x = "Disease Status", y = "Serum Cholesterol (mg/dl)",
       title = "Distribution of Serum Cholesterol by Disease Status") +
  theme_minimal() +
  theme(legend.position = "none")
```
```{r}
x_chol = chol_df$chol[chol_df$disease_status == "Diseased"]
y_chol = chol_df$chol[chol_df$disease_status == "Not Diseased"]
sd_ratio_chol = max(sd(x_chol), sd(y_chol)) / min(sd(x_chol), sd(y_chol))
round(c(sd(x_chol), sd(y_chol), sd_ratio_chol), 2)
```

```{r}
# Normality check
qq_chol = ggplot(chol_df, aes(sample = chol)) +
  stat_qq(col = "blue") +
  stat_qq_line(col = "red") +
  facet_wrap(~disease_status) +
  theme_minimal() +
  labs(title = "Normal QQ Plots of Cholesterol by Disease Status",
       x = "Theoretical quantiles", y = "Sample quantiles (chol)")

qq_chol
```
```{r}
hist_chol = ggplot(chol_df, aes(x = chol)) +
  geom_histogram(bins = 40, fill = "green", colour = "blue") +
  facet_wrap(~disease_status) +
  theme_minimal() +
  labs(title = "Histograms of Cholesterol by Disease Status",
       x = "Serum Cholesterol (mg/dl)", y = "Count")
```
```

hist_chol
```
```{r}
# Two-sample t-test (pooled; use var.equal = FALSE for Welch)
t.test(chol ~ disease_status, data = chol_df, var.equal = TRUE)
```
```{r}
# Bootstrap
N_bootstraps = 5000
boot_diff_chol = numeric(N_bootstraps)
set.seed(42)
for (i in 1:N_bootstraps) {
  x_sample = sample(x_chol, length(x_chol), replace = TRUE)
  y_sample = sample(y_chol, length(y_chol), replace = TRUE)
  boot_diff_chol[i] = mean(x_sample) - mean(y_sample)
}
boot_chol_df = data.frame(boot_diff_chol)

# 95% percentile bootstrap CI (Diseased - Not Diseased)
boot_ci_chol = quantile(boot_diff_chol, c(0.025, 0.975))
boot_ci_chol
```
```{r}
# Bootstrap distribution
ggplot(boot_chol_df, aes(x = boot_diff_chol, y = after_stat(density))) +
  geom_histogram(fill = "green", col = "blue") +
  geom_vline(xintercept = boot_ci_chol, linetype = "dashed") +
  labs(title = "Bootstrap distribution of mean difference (chol)",
       x = "Difference in means (mg/dl)", y = "Density")

# Comparison table of intervals
t_ci_chol = t.test(x_chol, y_chol, var.equal = TRUE)$conf.int
data.frame(method = c("Pooled t", "Bootstrap"),
           lower = c(t_ci_chol[1], boot_ci_chol[1]),
           upper = c(t_ci_chol[2], boot_ci_chol[2]))

```
Exercise Induced Angina (Exang):

```{r}
#This code is used to first turn 1 and 0s to diseased and non diseased (for better readability)
#And also True=Angina, False=No Angina.
exang_df = health_data_df %>%
  filter(!is.na(exang)) %>%
  mutate(disease_status = ifelse(diseased == 1, "Diseased", "Not Diseased"),
         exang_status = ifelse(exang == TRUE, "Angina", "No Angina"))

#This was picked up from DATA 601, python code converted to R to check how much missing
information per origin AND how much % patients have heart disease, in each location

```
exang_missing = health_data_df %>%
  group_by(origin) %>%
  summarise(n = n(),
            n_missing_exang = sum(is.na(exang)),
            pct_diseased = round(100 * mean(diseased), 1))
exang_missing
```
```{r}
#This was run to just view the number of healthy and diseased people with and without angina
exang_table = table(exang_df$disease_status, exang_df$exang_status)
exang_table
```
```{r}
#This code splits the data into diseased and non diseased. Then read each line (comment in front)
exang_summary = exang_df %>%
  group_by(disease_status) %>%
  summarise(n = n(),  # count total
            n_angina = sum(exang),   #count cases where exang=TRUE
            n_no_angina = sum(!exang), #Count cases where exang=FALSE (thats why !)
            prop_angina = round(mean(exang), 4)) #mean of 0s and 1, round off to 4 figures
exang_summary
```
```{r}
# AI helped to develop and understand this code to visualize the % of angina and no angina in each of
the two groups (diseased vs no diseased)
bar_exang = ggplot(exang_df, aes(x = disease_status, fill = exang_status)) +
  geom_bar(position = "fill", width = 0.6) +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Disease Status", y = "Percentage of Patients",
       fill = "Exercise-Induced Angina",
       title = "Exercise-Induced Angina by Disease Status") +
  theme_minimal()

bar_exang
```
```{r}
# Split exang into the two groups (TRUE = angina)
angina_diseased = exang_df$exang[ exan g_d f disease_status == "Diseased"]
angina_not_diseased = exang_df$exang[ exan g_d f disease_status == "Not Diseased"]
```
```{r}
# Two-proportion test (Diseased - Not Diseased)
prop_test_exang = prop.test(x = c(sum(angina_diseased), sum(angina_not_diseased)),
                            n = c(length(angina_diseased), length(angina_not_diseased)),
                            alternative = "two.sided", conf.level = 0.95, correct = FALSE)
prop_test_exang

z_exang = sqrt(prop_test_exang$statistic)
z_exang
```
```{r}
```

```
# Bootstrap
N = 5000
boot_diff_exang = numeric(N_bootstraps)
set.seed(42)
for (i in 1:N) {
  diseased_sample = sample(angina_diseased, length(angina_diseased), replace = TRUE)
  not_diseased_sample = sample(angina_not_diseased, length(angina_not_diseased), replace = TRUE)
  boot_diff_exang[i] = mean(diseased_sample) - mean(not_diseased_sample)
}
boot_exang_df = data.frame(boot_diff_exang)

boot_ci_exang = quantile(boot_diff_exang, c(0.025, 0.975))
boot_ci_exang
```
```{r}

ggplot(boot_exang_df, aes(x = boot_diff_exang, y = after_stat(density))) +
  geom_histogram(fill = "green", col = "blue") +
  geom_vline(xintercept = boot_ci_exang, linetype = "dashed") +
  labs(title = "Bootstrap distribution of difference in proportions (exang)",
       x = "Difference in proportions (Diseased - Not Diseased)", y = "Density")
```
```{r}
# Comparison table of intervals
prop_ci_exang = prop_test_exang$conf.int
ci_compare_exang = data.frame(method = c("prop.test", "Bootstrap"),
                              estimate = c(diff_exang, diff_exang),
                              lower = c(prop_ci_exang[1], boot_ci_exang[1]),
                              upper = c(prop_ci_exang[2], boot_ci_exang[2]))
ci_compare_exang
```
```{r}
# Dot-and-whisker plot of the two intervals
ggplot(ci_compare_exang, aes(x = estimate, y = method)) +
  geom_errorbar(aes(xmin = lower, xmax = upper), width = 0.15, colour = "blue",
                orientation = "y") +
  geom_point(size = 3, colour = "red") +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(title = "95% CIs for Difference in Proportion with Angina",
       x = "Difference in proportions (Diseased - Not Diseased)", y = "Method") +
  theme_minimal()
```